

Family-wise error rate (FWER)

It is the probability of making at least one Type I error (false positive) when conducting multiple statistical tests simultaneously.

- **Alpha inflation:** When running multiple independent tests, the probability of at least one false positive grows quickly. With 20 tests at $\alpha = 0.05$, the chance of a false positive is $1 - (1 - 0.05)^{20} \approx 64\%$, far higher than 5%

Formula

$$FWER = 1 - (1 - \alpha)^m$$

where:

- α = significance level of individual tests
- m = the number of tests

Why FWER?

- Example: In clinical trials, testing multiple treatments simultaneously increases the chance of finding at least one falsely significant result.

Method to correct the FWER

1. Bonferroni Correction

The Bonferroni correction is the simplest and most conservative method. It adjusts the significance level for each test by dividing the desired overall significance level α by the number of tests m :

$$\alpha_{\text{adjusted}} = \frac{\alpha}{m}$$

Example:

If we conduct 10 hypothesis tests with a significance level of $\alpha = 0.05$:

$$\alpha_{\text{adjusted}} = \frac{0.05}{10} = 0.005$$

The Bonferroni correction reduces the risk of false positives but can be overly conservative, reducing the power of the test.

2. Holm-Bonferroni Method

The Holm-Bonferroni method is a step-down procedure that is less conservative than Bonferroni. It sorts the p -values in ascending order and applies the following formula:

$$\alpha_{\text{adjusted}} = \frac{\alpha}{m - (k - 1)}$$

Where: k is the rank of the hypothesis.

Steps:

- Rank the p -values from smallest to largest.
- Apply the adjusted threshold incrementally.

3. Šidák Correction

The Šidák correction assumes the tests are independent. It adjusts the significance level using the formula:

$$\alpha_{\text{adjusted}} = 1 - (1 - \alpha)^{1/m}$$

For small α values, the Šidák correction is similar to the Bonferroni correction but slightly less conservative.

4. Benjamini-Hochberg (False Discovery Rate)

The Benjamini-Hochberg (BH) method controls the False Discovery Rate (FDR) rather than the FWER. It ranks the p -values in ascending order and compares each p -value to:

$$\frac{k}{m} \times \alpha$$

Where:

- k is the rank.
- m is the total number of tests.

The BH method is more powerful than FWER corrections but allows for a small proportion of false positives.